Elements of a Finite Abelian Group

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Group Theory Problems

Problem 1: Finite Abelian Group Order

Let G be a finite Abelian group of cardinality k. Show that for each $g \in G$, $g^k = e$.

The Proof

Because the group is Abelian, the order in which we multiply the elements of G does not matter.

Let the elements of our group be $G = \{g_1, g_2, \ldots, g_k\}$.

Now, fix an arbitrary element $g \in G$. If we multiply every element in G by this g, we get a new set: $gG = \{gg_1, gg_2, \ldots, gg_k\}$. Since left-multiplication by a group element is a bijection (a permutation of the group elements), this new set gG is exactly the same as the original set G.

Since the sets are identical and G is Abelian, taking the product of all elements in gG must equal the product of all elements in G:

$$\prod_{i=1}^{k}(gg_i) = \prod_{i=1}^{k} g_i$$

Because the group is Abelian, we can pull the g's to the front:

$$g^k(g_1 g_2 \cdots g_k) = (g_1 g_2 \cdots g_k)$$

Since $(g_1 g_2 \cdots g_k)$ is just an element of the group, we can multiply both sides by its inverse to cancel it out. This leaves us with:

$$g^k = e$$

And the proof is complete. ■